Financing the Air We Breathe: Climate vs. Non-Climate Aid and Local CO2 Emissions[*]

Staggered Event-Study and Dose–Response Evidence from Global ADM2 Data

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Abstract

This paper links geocoded development projects to subnational (ADM2) CO outcomes from EDGAR for 2000–2022 to study whether climate-tagged aid lowers emissions where it is spent. I implement a staggered difference-in-differences design with not-yet-treated controls (Callaway–Sant'Anna) to estimate dynamic effects of first project arrival and then quantify heterogeneity by cumulative post-treatment exposure (dose–response). Pre-treatment trends are flat. Climate portfolios yield modest but significant medium-run reductions in local log-CO , and these declines intensify with larger cumulative climate funding. By contrast, non-climate aid slightly raises emissions, consistent with general production scale effects. The results speak to aid effectiveness and targeting: (i) scaling climate-tagged portfolios can deliver local decarbonization; (ii) steering non-climate resources away from carbon-intensive uses may avoid offsetting impacts. The paper provides micro-spatial evidence on how the composition and intensity of external finance shape environmental outcomes in developing economies.

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Introduction

International and domestic climate finance increasingly flows through, or must be implemented by, sub-nationalgovernments. Cities and regions control large shares of infrastructure investment, land-use regulation, and service delivery. Yet credible evidence on **who receives climate finance, at what cost, and with what effects** remains fragmented across disciplines. This article organizes that evidence articulates a tractable conceptual framework, and maps it to empirical strategies suitable for causal evaluation.

Literature Review

The relationship between development, climate aid/finance, and CO2 emissions is complex. Climate finance and targeted aid can reduce CO2 emissions, but the effectiveness depends on the type of aid, governance, and the stage of economic development. Economic growth and financial development alone often increase emissions unless paired with green finance or strong environmental policies.

Climate finance, especially when directed toward mitigation, is associated with lower CO2 emissions, with stronger effects in countries with higher economic development and in small island states (C.-C. Lee et al. 2022; Khan, Ahmed, and Hull 2023). Green finance and green technology innovation consistently show significant negative impacts on CO2 emissions in both developed (G7, OECD) and emerging economies (BRICS) (Sharif et al. 2022; Saeed Meo and Karim 2022; Umar and Safi 2023; Sadiq et al. 2024). Climate aid for renewable energy has a slight but transitory effect on reducing CO2 emissions, with greater impact after a certain threshold. However, its effectiveness is limited if recipient countries prioritize economic over ecological efficiency (Kablan and Chouard 2022).

Regarding Official Development Assistance (ODA), it can directly and indirectly mitigate CO2 emissions, but the effect is stronger in middle-income countries and when governance is robust. In low-income countries, the impact is limited (Barkat, Alsamara, and Mimouni 2024; S. K. Lee et al. 2020; Farooq 2021; Mahalik et al. 2021). Aid directed toward fossil fuel development increases CO2 emissions, especially in countries with less environmental regime integration (Henderson and Sommer 2022; Henderson 2019). Strong governance and environmental policies enhance the effectiveness of aid and finance in reducing emissions (Barkat, Alsamara, and Mimouni 2024; Farooq 2021; Henderson 2019).

Recent research provides clear evidence that green finance and climate aid can significantly reduce CO2 emissions at the subnational (regional or city) level, but the effectiveness varies by region, policy design, and local economic structure.

Studies using data from hundreds of Chinese cities and provinces show that green finance policies and instruments (such as green loans and subsidies) have a significant negative effect on CO2 emissions intensity at the subnational level. The effect is stronger in eastern and non-resource-based cities, while central and western or resource-dependent regions see weaker impacts (Y. Sun, Bao, and Taghizadeh-Hesary 2023; F. Wang, Cai, and Elahi 2021; Zhao et al. 2024; Li et al. 2025). Uniform national policies may be less effective than regionally tailored approaches. Incentive-based policies work better in developed eastern provinces, while a mix of supportive and pressure policies (like carbon taxes) are needed in less developed central regions (Y. Sun, Bao, and Taghizadeh-Hesary 2023; Li et al. 2025) Green finance reduces emissions by supporting green technology innovation, upgrading industrial structure, and promoting renewable energy adoption. Debt-based green finance instruments are more effective than equity-based ones in reducing emissions at the city level (F. Wang, Cai, and Elahi 2021; Zhao et al. 2024) . At the micro level, green finance helps firms reduce emissions through lower financing costs, technological innovation, and external supervision. However, emission reduction is less pronounced in heavy-polluting and resource-based industries (Zhao et al. 2024).

Climate finance projects, those specifically designed to support low-carbon, environmentally sustainable activities, consistently outperform traditional development finance projects in reducing CO2 emissions at the subnational (regional or city) level. Climate finance significantly reduces local and neighboring CO2 emissions, with a 1% increase in green financial development linked to a 3% decrease in regional emissions (J. Sun et al. 2022; Chen and Chen 2021; Su, Qiao, and Xu 2024). The effect is strongest in economically advanced or eastern regions and in non-resource-based cities, where financial markets and green policy implementation are more mature (Chen and Chen 2021; Su, Qiao, and Xu 2024). Climate finance also creates "ripple effects," reducing emissions in adjacent areas by promoting green technology and industrial upgrades (J. Sun et al. 2022; Chen and Chen 2021; Su, Qiao, and Xu 2024).

General development finance (not specifically climate-related) can have mixed or even adverse effects on emissions. In some cases, it increases emissions by fueling economic growth and energy consumption, especially in less developed or resource-dependent regions (Chen and Chen 2021; Su, Qiao, and Xu 2024; L. Wang, Yang, and Cai 2024; Mumuni and Hamadjoda Lefe 2023) (Chen & Chen, 2021; Su et al., 2024; Wang et al., 2023; Mumuni & Lefe, 2023). The emission-reducing effect of traditional finance is weaker, and in some regions, it may actually exacerbate pollution unless paired with strong environmental regulations or targeted toward green sectors (Chen & Chen, 2021; Wang et al., 2023). Traditional finance lacks these targeted mechanisms and may inadvertently support high-emission sectors (Chen & Chen, 2021; Wang et al., 2023).

At the subnational level, climate finance projects are significantly more effective than traditional development finance in reducing CO2 emissions, especially in regions with mature financial systems and strong policy support. Traditional finance may have limited or even negative environmental effects unless explicitly directed toward green objectives.

Taken together, the existing evidence shows that climate-oriented finance can curb CO emissions, especially where governance and market depth support absorption, yet most subnational findings are China-centric, many designs are correlational, and "climate" vs. "traditional" finance is often measured with coarse, donor-provided labels. This leaves three central gaps: (i) limited external validity at fine geographic scales, (ii) weak causal identification of dynamic, project-level effects and spillovers, and (iii) insufficient separation of mechanisms (mitigation vs. adaptation; sectoral channels) and measurement error in climate tagging.

This study addresses these gaps by assembling a global ADM2 panel that links project-level development and climate finance (Bomprezzi et al. 2024) to high-resolution emissions outcomes, and by using ClimateFinanceBERT (Toetzke, Stünzi, and Egli 2022) to validate climate content beyond donor labels. Leveraging modern staggered DiD estimators, I identify the dynamic effects of climate and traditional finance on CO levels and intensity. The project granularity enables clean contrasts between mitigation and adaptation portfolios, while text-audited tagging provides the first large-scale test of greenwashing/mislabeling in aid data. By delivering the first worldwide, causally identified, subnational assessment of how climate vs. traditional development finance shape emissions, this study moves the literature from suggestive correlations toward place-based evidence.

I leverage spatially explicit project data and a modern identification strategy to estimate the causal effect of climate finance on local pollution. This design combines (i) staggered adoption DiD (Callaway and Sant'Anna 2021) to identify average dynamic effects and (ii) dose–response analyses that compare dynamics across post-treatment exposure bins and via continuous doses at the ADM2–year level. This two-pronged approach distinguishes targeting from treatment intensity and is well-suited to the lumpy, time-varying nature of climate finance.

Data

I construct an ADM2–year panel by integrating two sources. First, I use the GODAD project database (Bomprezzi et al. 2024). In order to get know whether or not projects in this datset were climate related, I use ClimateFinanceBERT (Toetzke, Stünzi, and Egli 2022) to classify projects from GODAD into climate-relevant or not and for those that were climate relevant, into adaptation, mitigation or environment categories. It allows me to get a description-based estimation of overall climate finance. For financing, I prioritize disb_loc_evensplit and substitute comm_loc_evensplit when the former is unavailable. Second, I merge environmental outcomes from EDGAR, aggregated to the ADM2 level by area-weighted averaging.

For each category, the adoption year for ADM2 j is defined as the first year with a strictly positive monetary amount:

$g_j \equiv \min\{t : \text{Amount}_{jt} > 0\}$.

Post-treatment exposure is the cumulative amount from the adoption year to the end of the panel:

$\text{Exposure}_j \equiv \sum_{t \geq g_j} \text{Amount}_{jt}$.

I partition treated ADM2s into intensity bins given by the empirical quantiles of Exposure_j. For descriptive cross-country figures, country–year totals are computed directly from GODAD by aggregating project amounts to the country–year level.

Empirical Strategy

Let Y_{jt} denote log pollution in ADM2 j and year t. We estimate group-time average treatment effects using the **CS** estimator with not-yet-treated controls:

$$ATT(g,e) = \mathbb{E}[Y_{jt}(1) - Y_{jt}(0) \mid g_j = g,, t - g = e],$$

and aggregate to dynamic event-time profiles with simultaneous confidence bands. To study heterogeneity by dose, we compute post-treatment totals (commitments/disbursements) and re-estimate dynamics within dose bins. This recovers a policy-relevant "more money means larger effects?" gradient while preserving the staggered-adoption identification.

Sample description

Table 1: Panel overview (ADM2–year).

Statistic	Value
Obs. (ADM2–year)	878 669
ADM2 units	38 203
Years	23
Year span	2000–2022
Ever treated ADM2	4 736
Never treated ADM2	33 467
Share treated by last year	12.4%

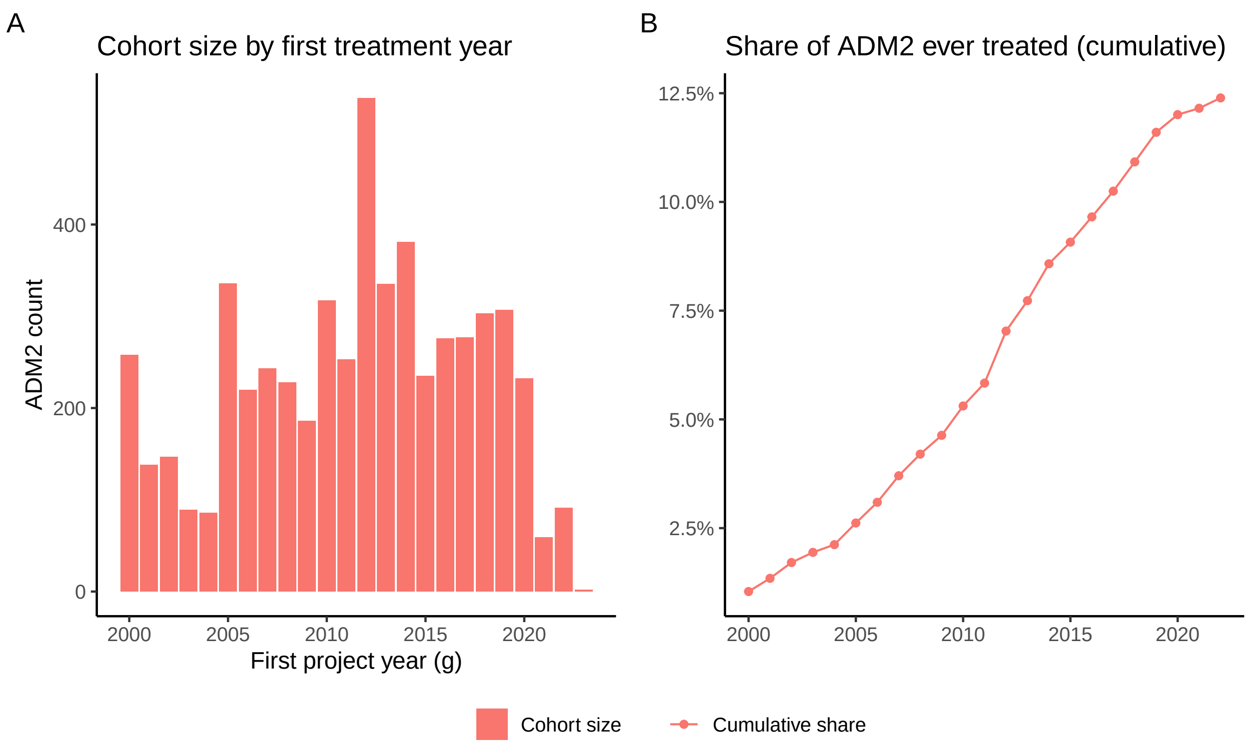
One can observe from Table 1 that the panel is large and nearly balanced, 38,203 ADM2 units observed over 23 years (2000–2022) yielding 878,669 unit–years, so missingness is minimal and pre-trend and medium-run dynamics are feasible. Only 12.4% of ADM2s are ever treated by 2022 (4,736 units), which provides abundant never- or not-yet-treated controls well suited to modern staggered DiD designs, while warranting overlap diagnostics and, if necessary, reweighting or trimming to common support. Staggered adoption and the definition of treatment as the first positive climate amount naturally motivate dynamic event-study ATTs alongside dose–response analyses based on cumulative post-treatment exposure. For clarity, note that the reported "share treated by last year" is the fraction ever treated by the final year, not the fraction treated in that year. Given the sample size and number of treated clusters, average effects should be precisely estimated.

Table 2: Descriptive statistics for log CO_2 outcomes: overall and by treatment status.

Group	N	Mean	Sd	P10.10%	P50.50%	P90.90%
Overall	**878669.000**	**10.558**	**2.072**	**8.308**	**10.412**	**13.195**
Never treated	769741.000	10.441	2.008	8.264	10.312	12.983

Ever treated	108928.000	11.386	2.312	8.744	11.148	14.511

In terms of outcomes, Table 2 shows that the distribution of log CO is clearly right-shifted for ever-treated ADM2s relative to never-treated ones: the mean gap is 0.945 log points, implying treated ADM2s exhibit about 2.57× higher CO levels on average $e^{0.945} \approx 2.57$, with a median gap of 0.836 log points $\approx$ 2.31×. Dispersion is larger among treated units (SD 2.31 vs 2.01), and the displayed percentiles are uniformly higher ($P10 : +0.48 \quad log \approx +62\% \quad P90 : +0.212 log \approx +24\%$), suggesting a broad rightward shift rather than just upper-tail differences. Cross-sectionally this is consistent with selection: treated ADM2s are likely more urban/industrial or otherwise systematically different. A pooled standardized mean difference of $\approx$ 0.46 (using group SDs and sample sizes) indicates moderate imbalance that should be addressed. Substantively, these gaps underscore the need to rely on within-ADM2 identification (unit and year fixed effects), probe parallel trends with event-study leads.



Note: g denotes the first year an ADM2 receives a positive climate amount (adaptation or mitigation). Panel A shows the count of ADM2s by g (years >= 2000).
Panel B plots the cumulative share of all ADM2s with g <= year; the denominator includes never−treated ADM2s.
Percent axis in Panel B refers to this share. Missing g implies never−treated; cohorts are right−censored at the panel end.

Figure 1: Adoption dynamics at ADM2 level: (A) cohort size by first treatment year; (B) cumulative share of ADM2 ever treated.

According to Figure 1, the adoption profile is clearly staggered and concentrated in the early-to-mid 2010s: Panel A shows modest early cohorts, a marked ramp-up around 2011–2016, and thinning cohorts thereafter (partly right-censoring near the panel end), while Panel B's S-shaped curve plateaus at $\approx$ 12–13% of ADM2s by 2022. Combined with earlier descriptives, treated ADM2s are systematically more CO -intensive, this pattern is consistent with selective rollout toward higher-emitting, more urban/industrial areas rather than random diffusion. Empirically, these features favor staggered event-study designs with rich fixed effects, but they also call for explicit pre-trend tests (anticipation and dose–response specifications using post-treatment exposure.

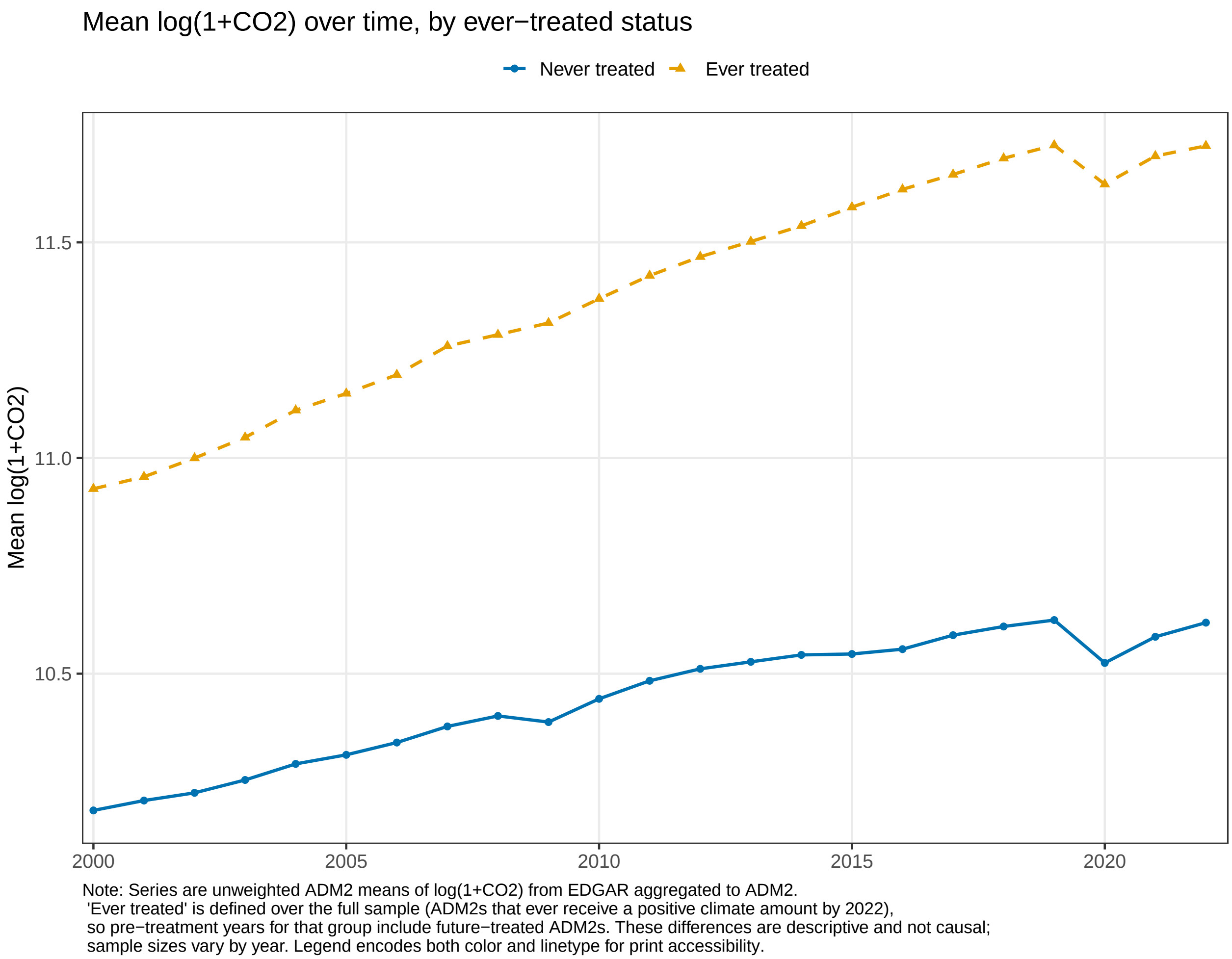


Figure 2: Unadjusted means of log(1+CO_2) by treatment status over time.

In Figure 1, the unadjusted series show a stable level gap, ADM2s that ever receive climate projects have consistently higher $log(1 + CO_2)$, but the two groups move along very similar secular paths. From 2000 to the early-mid 2010s both lines trend upward with comparable slopes, and there is no obvious pre-period divergence, which is reassuring for the parallel-trends requirement. Because "ever treated" is defined over the whole sample, these differences are descriptive, not causal, and reflect composition (future-treated units are included in pre-years). Accordingly, my identification will rely on within-ADM2 variation and dynamic event-study leads to test for anticipation. I will also probe heterogeneity and spillovers, but the aggregate pattern here is broadly consistent with a DiD design: different levels, similar trends.

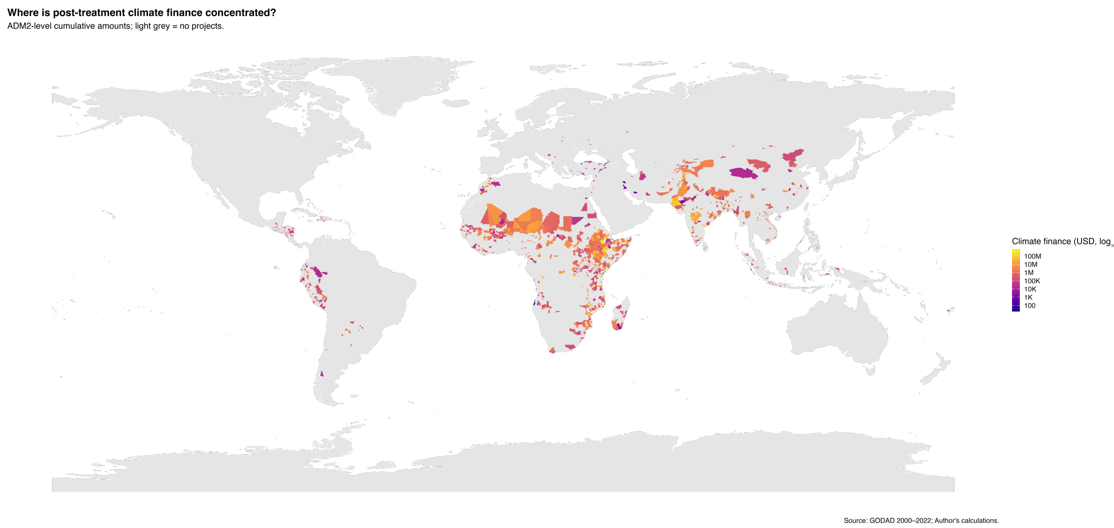


Figure 3: ADM2-level cumulative climate finance (USD, log scale). Light grey = no projects.

Figure 3 shows a highly clustered footprint of climate-project exposure at the ADM2 level with a dense concentrations across Sub-Saharan Africa (Sahel, Great Lakes, coastal West Africa), South Asia (Indo-Gangetic plain and Himalayan foothills), and parts of Southeast Asia, with smaller pockets in the Andes and Central America; coverage is sparse in North Africa/Middle East and essentially absent in high-income countries (by design). This spatial pattern is consistent with targeting toward populous/urbanizing and vulnerable corridors.

Results

We identify the **first year** in which an ADM2 receives at least one **climate** project. If the GODAD file already has an ADM2 code, we use it; if not, we perform a spatial join using the point geometry.

The Figure 4 dynamic event-study estimates (uniform 95% bands) indicate no evidence of anticipatory behavior: pre-treatment coefficients are small and statistically indistinguishable from zero across all specifications, supporting the parallel-trends assumption. Aggregating categories, **all climate** projects yield a modest negative long-run effect, whereas **non-climate** aid shows small positive effects, in line with the scale effects of general development

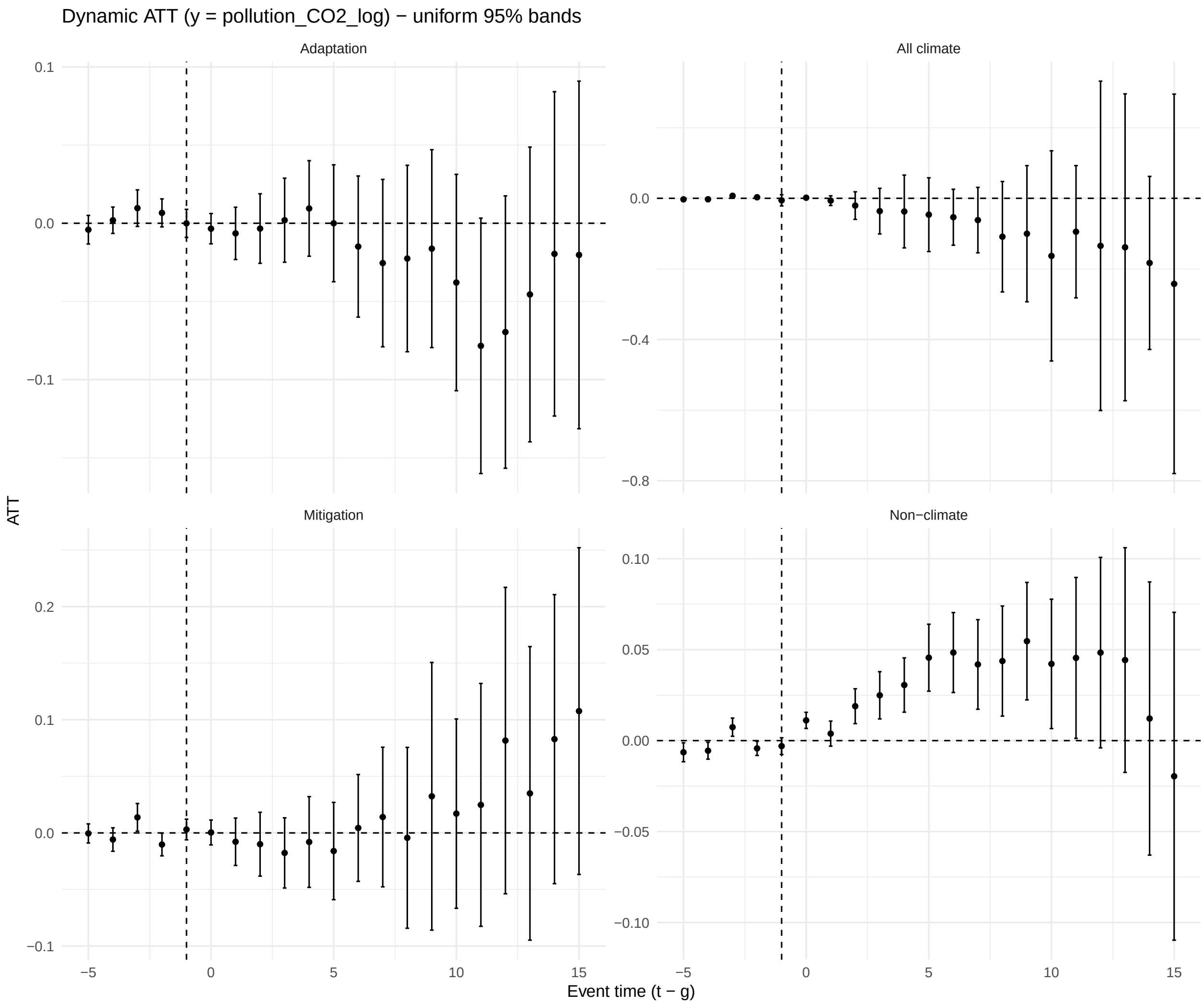


Figure 4: Dynamic treatment effects of aid adoption on local CO2 emissions (log). Event-time estimates (t−g) from Callaway–Sant'Anna DiD using not-yet-treated controls, doubly-robust estimator, and uniform 95% simultaneous bands (multiplier bootstrap; clustered at ADM2).

spending.

Intensity: event study by post-treatment dose bins

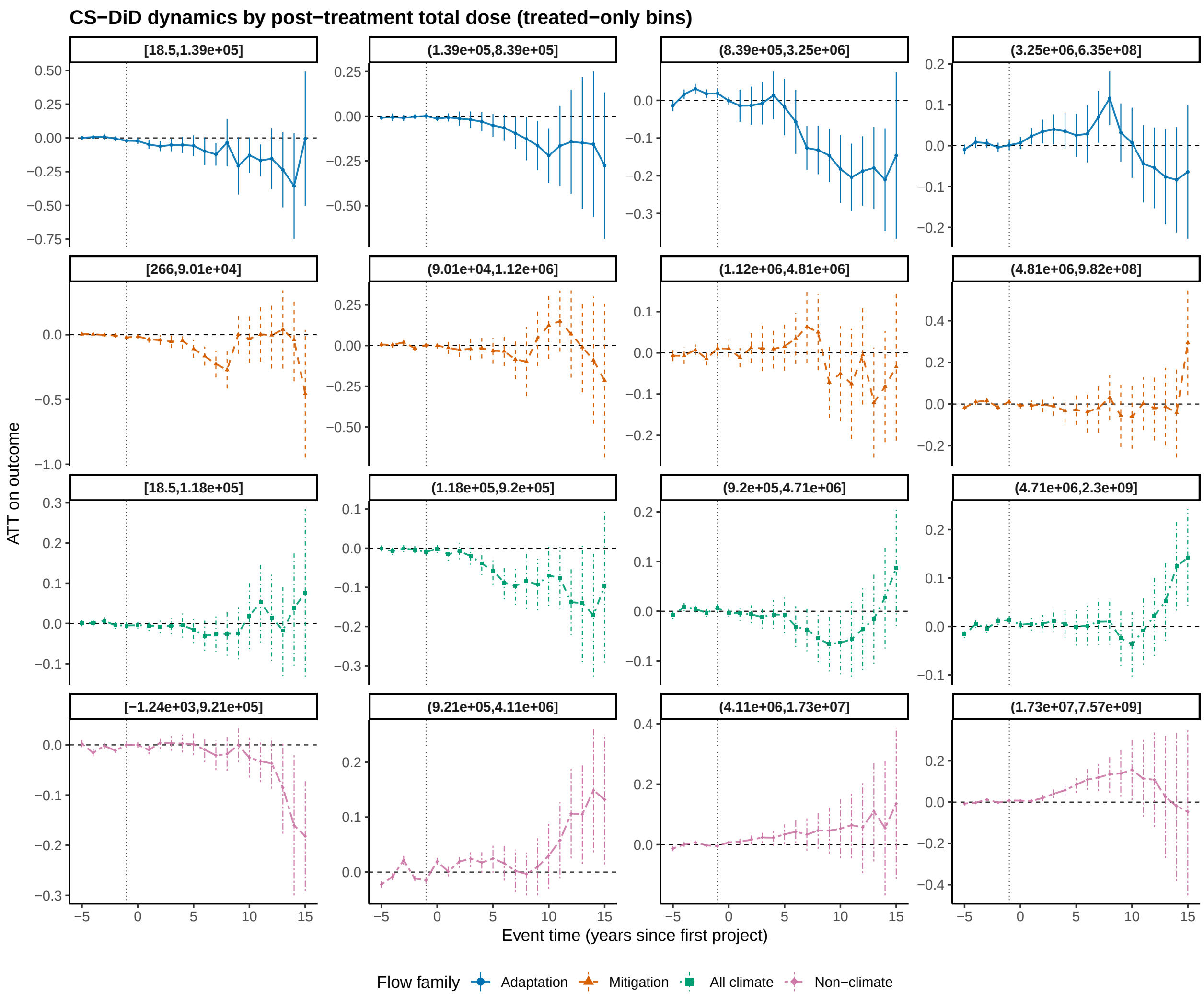


Figure 5: Event-study by post-treatment total dose — stratified bins (treated-only cuts).

Interpretation. Higher post-treatment exposure is associated with larger post-event estimates, consistent with a **dose–response** relationship. Pre-treatment coefficients remain centered at zero across bins.

Excluding China

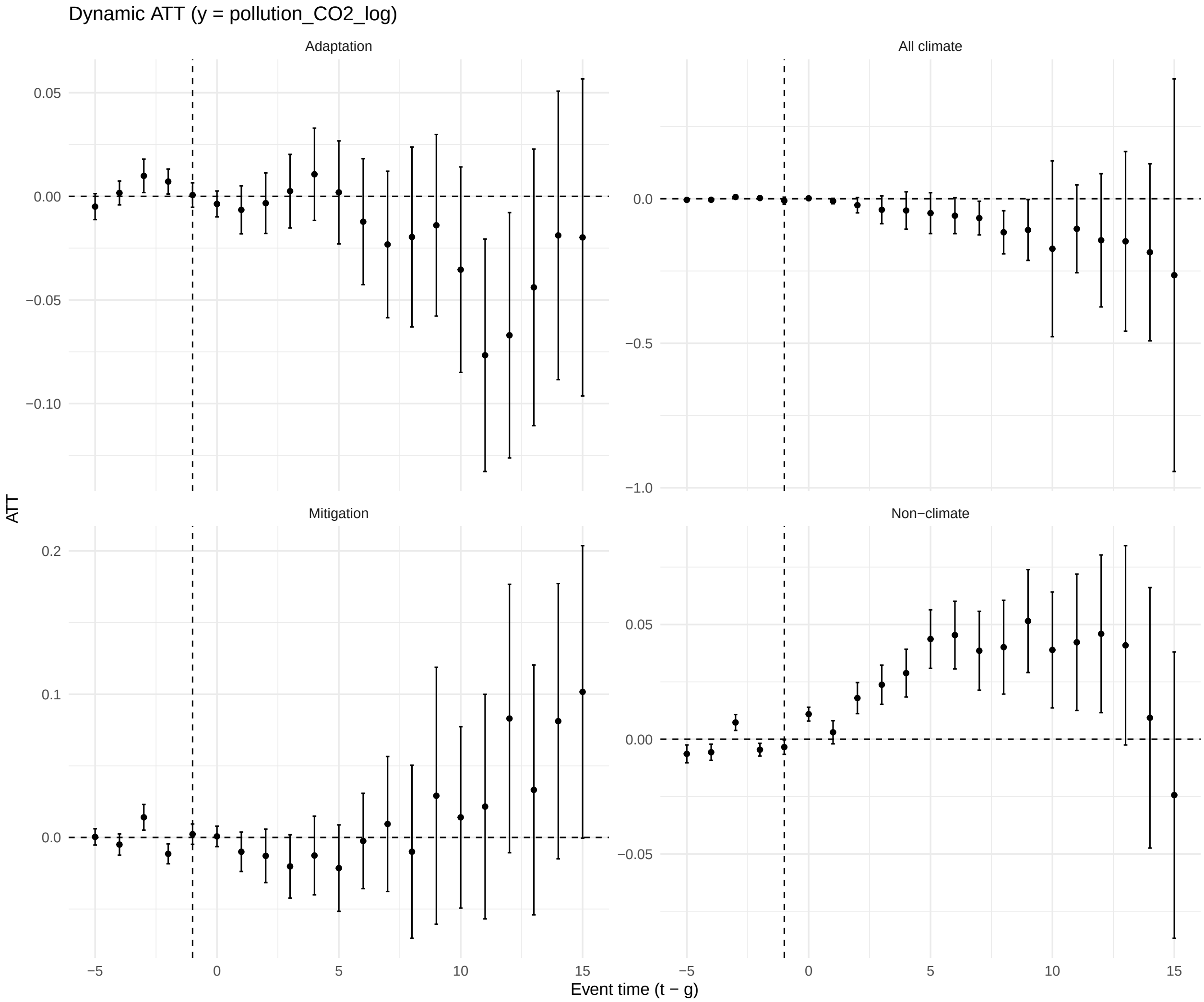
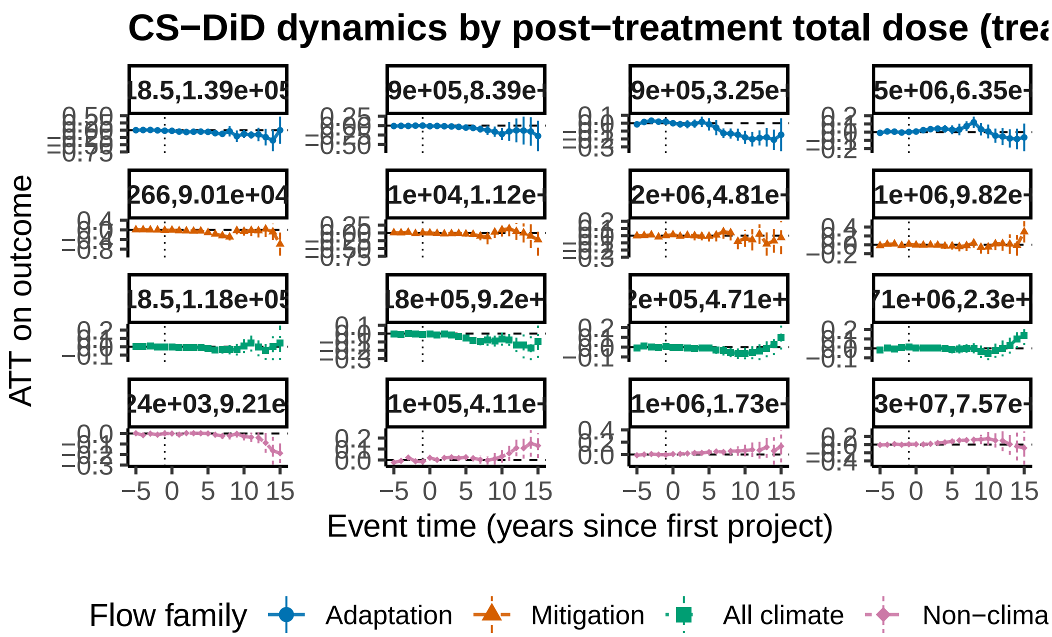
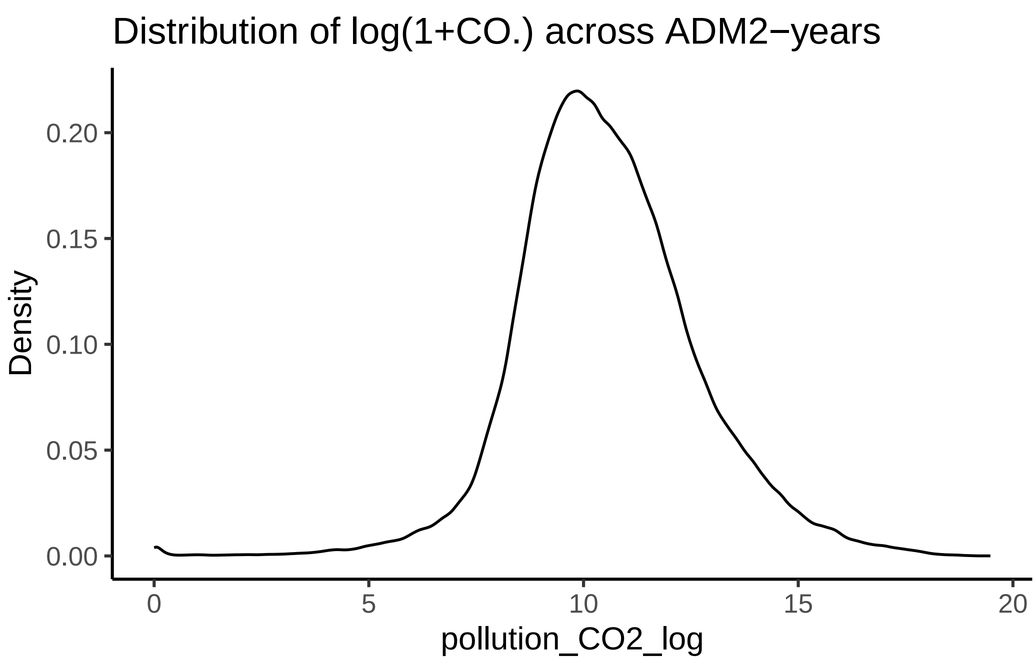
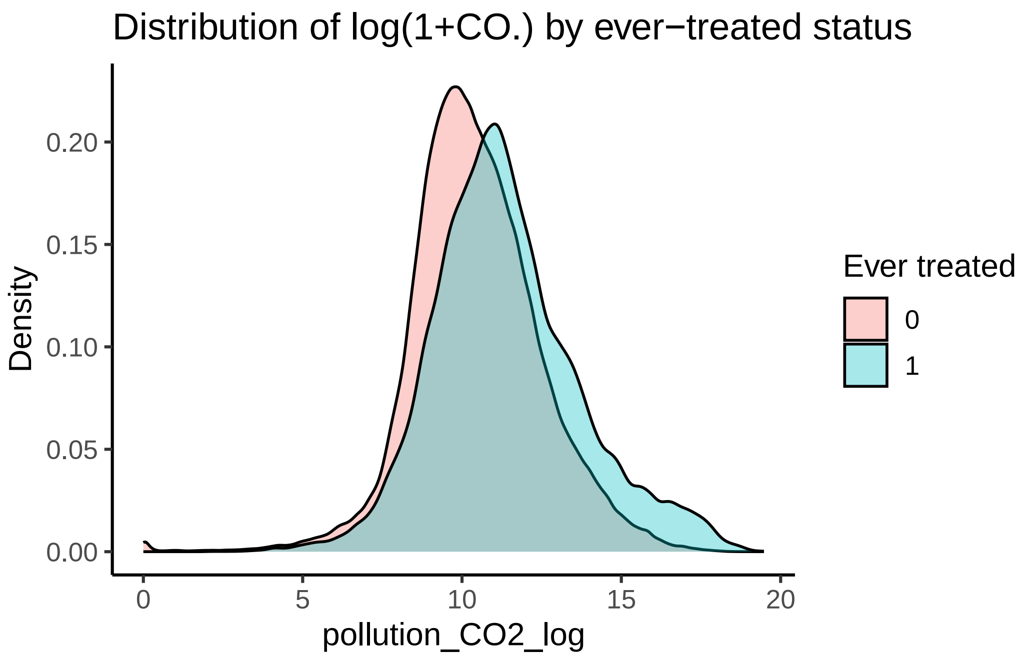


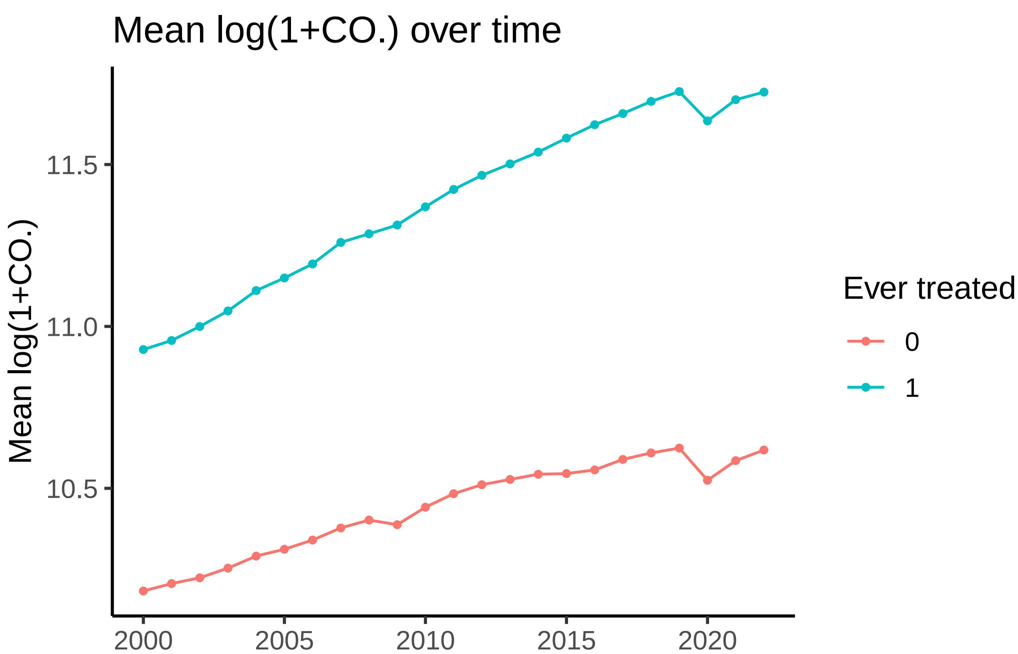
Figure 6: Dynamic treatment effects of aid adoption on local CO2 emissions (log, China excluded)

Intensity Adapt excluding China






Distribution of log(1+CO.) by ever−treated status
Density
0.00
0.05
0.10
0.15
0.20
0
5
10
15
20
pollution_CO2_log
Ever treated
0
1


Mean log(1+CO.) over time
Mean log(1+CO.)
10.5
11.0
11.5
2000
2005
2010
2015
2020
Ever treated
0
1

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Appendix

Top emitters (ADM2) — average over the sample window

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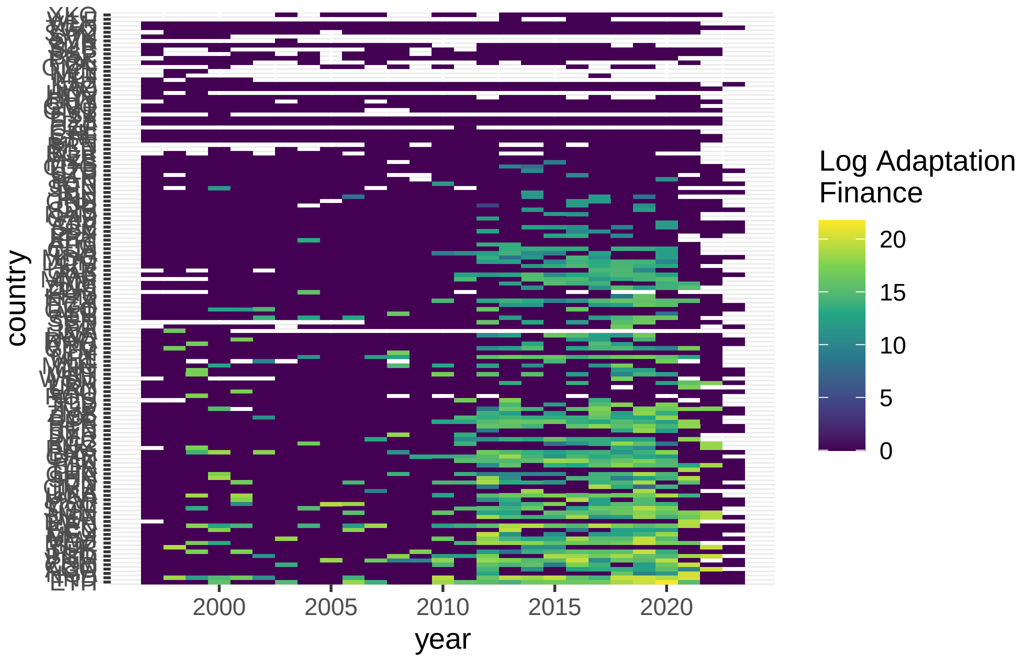
    <char>      <char>     <num>
 1:    CHN  CHN.10.9_1 199632759
 2:    CHN  CHN.24.1_1 141872754
 3:    CHN  CHN.15.7_1 133343969
 4:    CHN   CHN.3.1_1 120020252
 5:    ZAF   ZAF.6.3_1 115805012
 6:    CHN CHN.15.10_1 113811236
 7:    CHN CHN.23.13_1 104862150
 8:    CHN  CHN.27.1_1 103411029
 9:    CHN  CHN.10.4_1  99511528
10:    CHN CHN.23.17_1  98720889
11:    CHN  CHN.19.7_1  96086514
12:    CHN  CHN.15.4_1  87744994
13:    CHN CHN.23.16_1  83558272
14:    CHN CHN.12.16_1  78899516
15:    CHN  CHN.28.5_1  76522584
16:    CHN  CHN.23.9_1  74562077
17:    CHN CHN.13.12_1  72518799
18:    CHN  CHN.18.1_1  72069729
19:    CHN  CHN.13.4_1  70898769
20:    CHN  CHN.10.8_1  70602072
      iso3     adm2_id   avg_CO2
```

Heatmaps (country × year matrix)



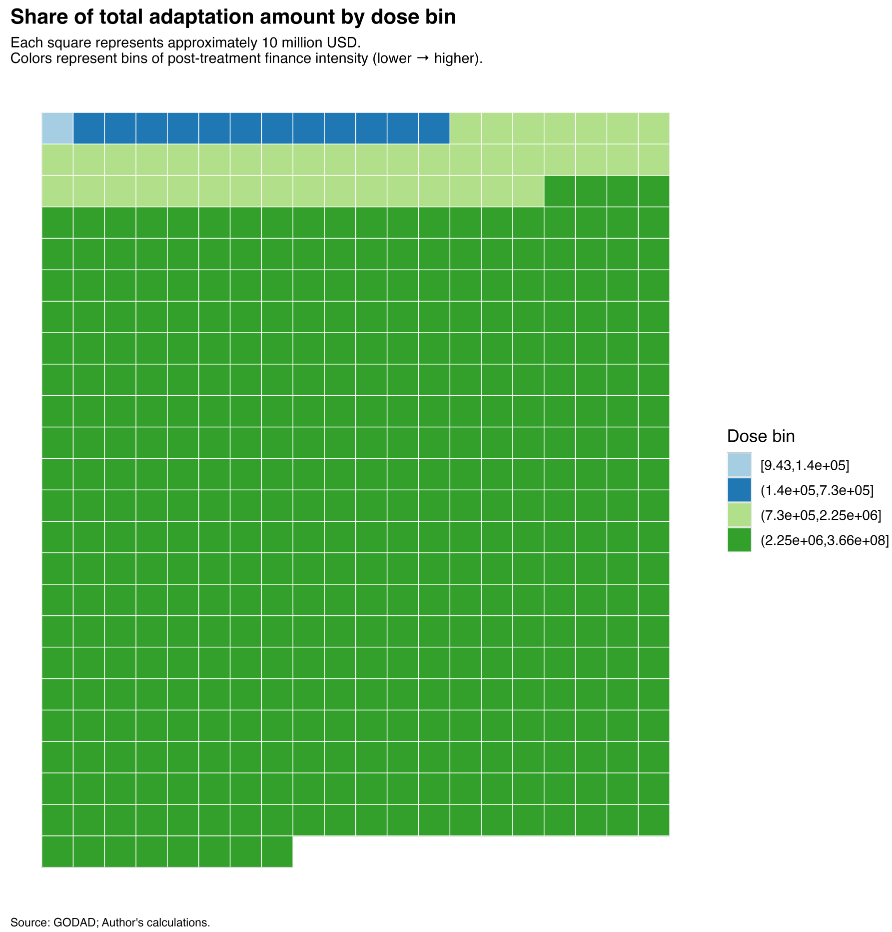


Figure A1: Adaptation bins post treatment intensity repartition

Reproducibility

```
R version 4.4.1 (2024-06-14)
Platform: aarch64-apple-darwin20
Running under: macOS 26.0.1

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dyli
LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dyli

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: Europe/Paris
tzcode source: internal

attached base packages:
```

```
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] forcats_1.0.0     patchwork_1.3.0   tibble_3.3.0      kableExtra_1.4.0
 [5] knitr_1.50        fixest_0.13.2     did_2.1.2         scales_1.4.0
 [9] ggplot2_3.5.2     janitor_2.2.1     stringr_1.5.1     sf_1.0-20
[13] arrow_21.0.0.1    readr_2.1.5       tidyr_1.3.1       dplyr_1.1.4
[17] data.table_1.17.0

loaded via a namespace (and not attached):
  [1] DBI_1.2.3            pROC_1.18.5          DRDID_1.2.0
  [4] sandwich_3.1-1       rlang_1.1.6          magrittr_2.0.3
  [7] dreamerr_1.4.0       snakecase_0.11.1     e1071_1.7-16
 [10] compiler_4.4.1       systemfonts_1.1.0    vctrs_0.6.5
 [13] reshape2_1.4.4       crayon_1.5.3         pkgconfig_2.0.3
 [16] fastmap_1.2.0        backports_1.5.0      labeling_0.4.3
 [19] rmarkdown_2.29       prodlim_2024.06.25   tzdb_0.5.0
 [22] tinytex_0.53         purrr_1.0.2          bit_4.6.0
 [25] xfun_0.53            trust_0.1-8          jsonlite_2.0.0
 [28] stringmagic_1.2.0    recipes_1.1.0        fastglm_0.0.3
 [31] uuid_1.2-1           broom_1.0.7          parallel_4.4.1
 [34] R6_2.6.1             stringi_1.8.7        RColorBrewer_1.1-3
 [37] parallelly_1.38.0    car_3.1-3            rpart_4.1.23
 [40] numDeriv_2016.8-1.1  lubridate_1.9.3      Rcpp_1.1.0
 [43] assertthat_0.2.1     iterators_1.0.14     BMisc_1.4.7
 [46] future.apply_1.11.3  zoo_1.8-13           Matrix_1.7-2
 [49] splines_4.4.1        nnet_7.3-19          timechange_0.3.0
 [52] tidyselect_1.2.1     abind_1.4-8          rstudioapi_0.17.0
 [55] dichromat_2.0-0.1    yaml_2.3.10          timeDate_4041.110
 [58] codetools_0.2-20     listenv_0.9.1        lattice_0.22-6
 [61] plyr_1.8.9           withr_3.0.2          evaluate_1.0.1
 [64] future_1.34.0        survival_3.7-0       units_0.8-7
 [67] proxy_0.4-27         xml2_1.3.8           pillar_1.11.0
 [70] ggpubr_0.6.0         carData_3.0-5        KernSmooth_2.23-24
 [73] foreach_1.5.2        stats4_4.4.1         generics_0.1.3
 [76] vroom_1.6.5          hms_1.1.3            globals_0.16.3
 [79] class_7.3-22         glue_1.8.0           tools_4.4.1
 [82] ModelMetrics_1.2.2.2 gower_1.0.2          ggsignif_0.6.4
 [85] grid_4.4.1           bigmemory_4.6.4      ipred_0.9-15
 [88] nlme_3.1-166         Formula_1.2-5        cli_3.6.5
 [91] bigmemory.sri_0.1.8  viridisLite_0.4.2    svglite_2.1.3
 [94] lava_1.8.1           gtable_0.3.6         rstatix_0.7.2
 [97] digest_0.6.37        classInt_0.4-11      caret_7.0-1
[100] farver_2.1.2         htmltools_0.5.8.1    lifecycle_1.0.4
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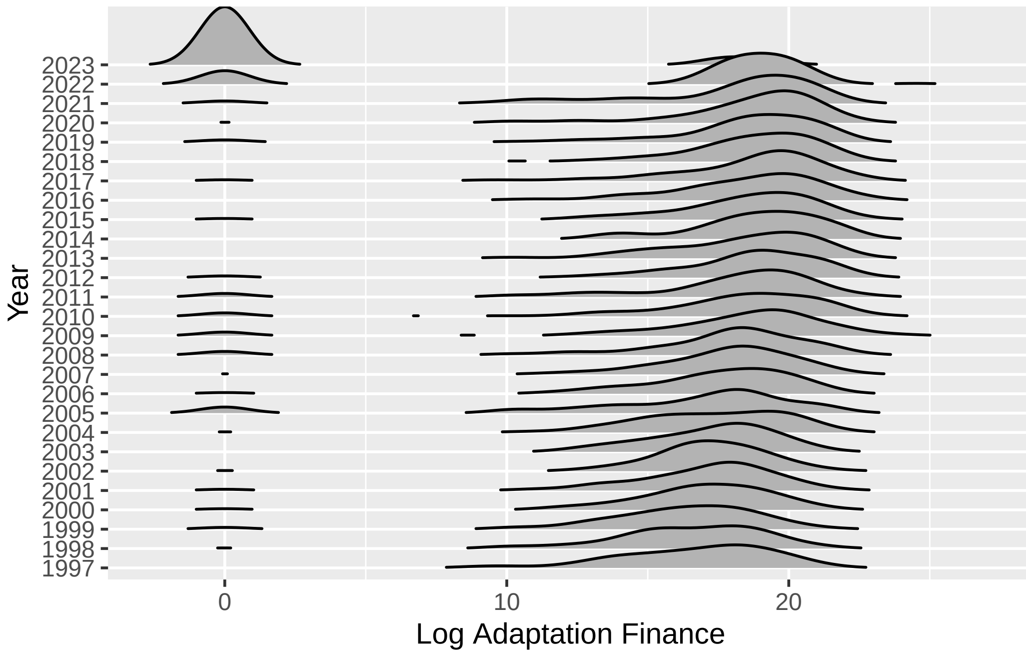
```


Year
2023
2022
2021
2020
2019
2018
2017
2016
2015
2014
2013
2012
2011
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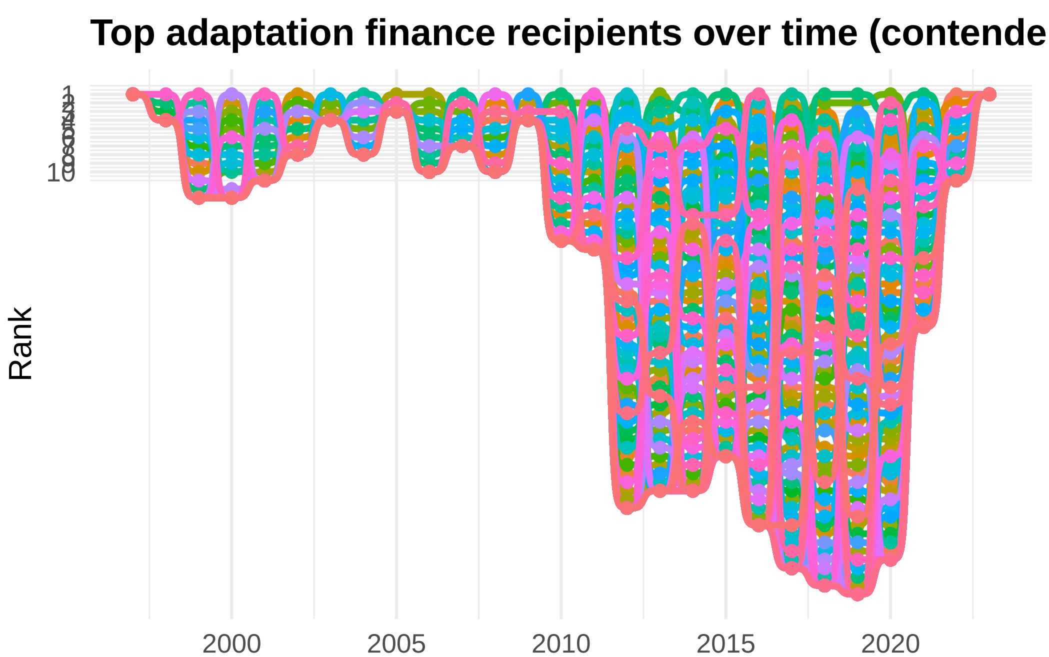
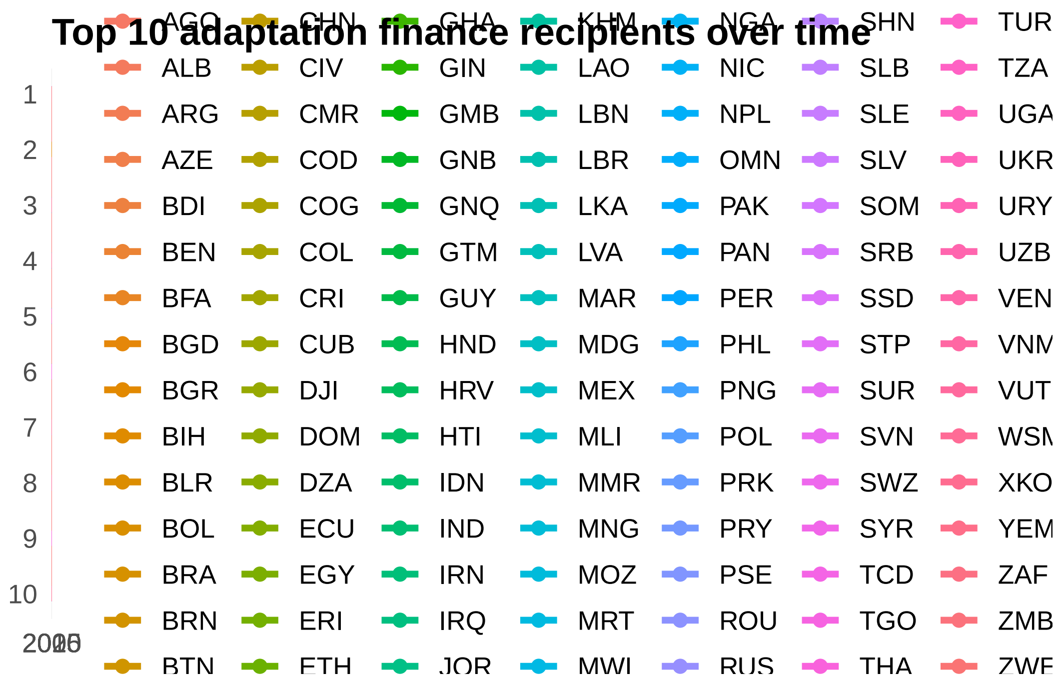
10
20
Log Adaptation Finance


Top adaptation finance recipients over time (contende
Rank
10
2000
2005
2010
2015
2020


Top 10 adaptation finance recipients over time
1
2
3
4
5
6
7
8
9
10
AGO
ALB
ARG
AZE
BDI
BEN
BFA
BGD
BGR
BIH
BLR
BOL
BRA
BRN
BTN
CHN
CIV
CMR
COD
COG
COL
CRI
CUB
DJI
DOM
DZA
ECU
EGY
ERI
ETH
GHA
GIN
GMB
GNB
GNQ
GTM
GUY
HND
HRV
HTI
IDN
IND
IRN
IRQ
JOR
KHM
LAO
LBN
LBR
LKA
LVA
MAR
MDG
MEX
MLI
MMR
MNG
MOZ
MRT
MWI
NGA
NIC
NPL
OMN
PAK
PAN
PER
PHL
PNG
POL
PRK
PRY
PSE
ROU
RUS
SHN
SLB
SLE
SLV
SOM
SRB
SSD
STP
SUR
SVN
SWZ
SYR
TCD
TGO
THA
TUR
TZA
UGA
UKR
URY
UZB
VEN
VUT
XKO
YEM
ZAF
ZMB